

Rami I. Hanna

linkedin.com/in/ramiihanna | rami-hanna.com

rami@rami-hanna.com | +1 (603) 293-3511

Boston, MA | U.S. Citizen

PROFILE

Robotics systems engineer with an M.Sc. in Autonomous Systems and a Summa Cum Laude B.S. in Electromechanical Engineering. Across **PerPlant**, **Harvard Microrobotics**, **Raytheon Technologies**, and **DEKA R&D**, built and debugged physical systems spanning ROS2, C++, Python, embedded systems, sensing, controls, computer vision, and validation. Strongest in multidisciplinary R&D where robotics, perception, software, hardware, and real-world constraints meet.

SKILLS

Programming / Systems: Python, C++, Embedded C, Bash, MATLAB, Linux/Ubuntu, Git, CMake

Robotics / Controls: ROS2, MicroROS, Gazebo, RViz/TF2, PID, Kalman/EKF, trajectory/path planning

Perception / ML: OpenCV, PyTorch, YOLOv8, scikit-learn, Open3D/PCL

Embedded / Sensing: NVIDIA Jetson, ESP32/STM32, CAN, UART/SPI/I2C, LiDAR, IMU, GNSS/GPS

Build / Test: Docker, GitHub Actions CI/CD, SOLIDWORKS (CSWA), LabVIEW, instrumentation

PROFESSIONAL EXPERIENCE



PerPlant | Robotics Co-op

02/25 - 11/25 | Copenhagen, DK

- Integrated ROS2 thermal/GPS field sensing in C++/Python using OpenCV, NVIDIA Jetson, and MicroROS; redesigned a multi-camera fixture to add thermal imaging and translated farmer feedback into technical requirements.
- Built CV/data curation tooling using ROI filtering, detector embeddings, UMAP/HDBSCAN, and grouped splitting to create representative annotation/evaluation batches from a 150,000+ image field dataset.



Harvard Microrobotics | Robotics Co-op

09/22 - 03/23 | Allston, MA

- Developed Flask/socket robot-operation interfaces spanning Embedded C and MicroROS for teleoperated/autonomous underwater workflows; also contributed to planetary-gearbox design and cross-subsystem debugging.



Raytheon Technologies | Innovation / Robotics Co-op

01/22 - 05/22 | Andover, MA

- Drove an epoxy-automation project from a manual-process problem to a working UR5e/PLC/Keyence vision system with modular CAD fixtures, gathering requirements across manufacturing and process stakeholders; the team estimated roughly \$7M in potential ROI.



DEKA Research and Development | Controls Intern

2019; 2020-2021 | Manchester, NH

- Built biomedical R&D test and traceability workflows spanning electrical/impact/thermal testing, PID-controlled LabVIEW/Modbus instrumentation, and RFID/Python/SQL data capture while working within FDA/clinical and patient-safety constraints.

SELECTED PROJECTS



M.Sc. Thesis | Tendon-Driven Finger Actuation

2026 | DTU

- Built an end-to-end tendon-driven finger-actuation research platform integrating reduced-order models, CAD/FDM hardware, wearable interfaces, motor/encoder control, and Python/OpenCV measurement.
- Validated 1,000/1,000 rigid-fixture cycles at 97.5% of a 49.93° engineering comparator with 0.26% mean period error; supervised three student interns across fixture, control, vision, and sensing work.



Teradyne Cartesian Robot | Connector Test System

2023 | Wentworth / Teradyne

- Built a Raspberry Pi/ESP32 Cartesian robot with load-cell feedback and automatic tool changing; achieved 0.01 mm travel precision and 0.5 g load standard deviation; published at IEEE SII/SICE 2024.



SunnySips | Founder & Full-Stack Developer

2025 - Present | Released iOS / Web Product

- Shipped and maintain a live iOS/web product using SwiftUI, FastAPI, Docker, automated testing, and GitHub Actions CI/CD; in-person café/user discovery changed feature priorities, including future-planning functionality.

EDUCATION



Technical University of Denmark (DTU)

2024-2026 | Lyngby, Denmark

M.Sc. Autonomous Systems | Thesis defended; requirements completed Aug. 2026
Full Merit Scholarship - one of 30 recipients



Wentworth Institute of Technology

2019-2023 | Boston, MA

B.S. Electromechanical Engineering
Summa Cum Laude · Dean's Award in Electromechanical Engineering